MECHANICAL FLOW DIAGRAM

DWG. NO.	TITLE
XB11A-0000-01	FLOW DIAGRAM AND EQUIPMENT INDEX
XB11A-0000-02	SYMBOLS AND IDENTIFICATIONS – PIPING
XB11A-0000-03	SYMBOLS AND IDENTIFICATIONS – INSTRUMENT
XB11A-0000-04	SYMBOLS AND IDENTIFICATIONS – EQUIPMENT
XB11A-0000-05	SYMBOLS AND IDENTIFICATIONS – INSTRUMENT CONNECTION DETAILS
XB11A-0000-06	SYMBOLS AND IDENTIFICATIONS – ELEC/INST/ PIPING
XB11A-0000-07	SYMBOLS AND IDENTIFICATIONS – EQ BRIDLE DRAIN
XB11A-0000-08	SYMBOLS AND IDENTIFICATIONS – PUMP SEAL PLAN
XB11A-0000-09	SYMBOLS AND IDENTIFICATIONS – PUMP SEAL PLAN
XB11A-0000-10	SYMBOLS AND IDENTIFICATIONS – SAMPLING TYPE
XB11A-0000-11	SYMBOLS AND IDENTIFICATIONS – SAMPLING SUMMARY
XB11A-1000-01A	FEED FILTERS (1/3)
XB11A-1000-01B	FEED FILTERS (2/3)
XB11A-1000-01C	FEED FILTERS (3/3)
XB11A-1000-02	FEED SURGE DRUM
XB11A-1000-03	FUEL GAS KO DRUM
XB11A-1000-04A	HEATER SECTION
XB11A-1000-04B	HEATER DECOKING SYSTEM
XB11A-1000-05	HDS REACTOR SECTION
XB11A-1000-06	POLYSHIFT REACTOR
XB11A-1000-07	RECYCLE GAS EXCHANGER
XB11A-1000-08	COLD SEPARATOR
XB11A-1000-09A	MAKE-UP GAS SECTION (1/3)
XB11A-1000-09B	MAKE-UP GAS SECTION (2/3)
XB11A-1000-09C	MAKE-UP GAS SECTION (3/3)
XB11A-1000-10A	RECYCLE GAS SECTION (1/3)
XB11A-1000-10B	RECYCLE GAS SECTION (2/3)
XB11A-1000-10C	ERCYCLE GAS SECTION (3/3)
XB11A-1000-11	HP LOOP EVACUATION
XB11A-1000-12	PRODUCT STRIPPER
XB11A-1000-13	PRODUCT STRIPPER REFLUX
XB11A-1000-14	NAPHTHA STABILIZER
XB11A-1000-15	NAPHTHA COOLING
XB11A-1000-16	PRODUCT COOLING
XB11A-1000-17	PRODUCT DRYING
XB11A-1000-18	RECYCLE GAS SCRUBBING
XB11A-1000-19	RICH AMINE FLASH DRUM
XB11A-1000-20	FUEL GAS SCRUBBING
XB11A-1000-21	LEAN AMINE SURGE DRUM
XB11A-1000-22	ANTIFOAMER INJECTION
XB11A-1000-23	SOUR WATER STRIPPING
XB11A-1000-24	SOUR WATER PUMP AROUND
XB11A-1000-25	STRIPPED SOUR WATER

UTILITY FLOW DIAGRAM

DWG. NO.	TITLE
XB12A-1000-01	STEAM & CONDENSATE SYSTEM (1/2)
XB12A-1000-02	STEAM & CONDENSATE SYSTEM (2/2)
XB12A-1000-03	DEAERATOR AND BFW PUMPS
XB12A-1000-04	BFW CHEMICAL DOSING
XB12A-1000-05	NITROGEN, PLANT AIR & INSTRUMENT AIR SYSTEM
XB12A-1000-06	FUEL GAS AND NATURAL GAS
XB12A-1000-07A	COOLING WATER & TEMPERED WATER (1/3)
XB12A-1000-07B	COOLING WATER & TEMPERED WATER (2/3)
XB12A-1000-07C	COOLING WATER & TEMPERED WATER (3/3)
XB12A-1000-08	DEMIN. AND UTILITY WATER SYSTEM
XB12A-1000-09	SOUR GAS FLARE SYSTEM
XB12A-1000-10	COMMON FLARE SYSTEM
XB12A-1000-11A	OIL DRAINAGE SYSTEM (1/2)
XB12A-1000-11B	OIL DRAINAGE SYSTEM (2/2)
XB12A-1000-12A	AMINE DRAINAGE SYSTEM (1/2)
XB12A-1000-12B	AMINE DRAINAGE SYSTEM (2/2)
XB12A-1000-13A	SOUR WATER DRAINAGE SYSTEM (1/2)
XB12A-1000-13B	SOUR WATER DRAINAGE SYSTEM (2/2)
XB12A-1000-14	UTILITY STATION AND EYE SHOWER

THE PIPING AND INSTRUMENT DIAGRAM INDEX

C-COMPRESSOR

EQUIPMENT	DWG. NO.
C-1201	XB11A-1000-10A
C-1301A	XB11A-1000-09A
C-1301B	XB11A-1000-09B

C-FAN & BLOWER

EQUIPMENT	DWG. NO.
C-1001A/B	XB11A-1000-04A
C-1002	XB11A-1000-04A

CT-STEAM TURINE

EQUIPMENT	DWG. NO.
CT-1201	XB11A-1000-10A

D-DRUM

EQUIPMENT	DWG. NO.
D-1001	XB11A-1000-02
D-1003	XB11A-1000-08
D-1004	XB12A-1000-03
D-1005	XB11A-1000-03
D-1006	XB11A-1000-04B
D-1007	XB11A-1000-01B
D-1008	XB11A-1000-01C
D-1009	XB11A-1000-01B
D-1010	XB12A-1000-10
D-1011	XB12A-1000-09
D-1015	XB12A-1000-07B
D-1101	XB11A-1000-13
D-1102	XB11A-1000-17
D-1201	XB11A-1000-10A
D-1301	XB11A-1000-09A
D-1302A	XB11A-1000-09A
D-1302B	XB11A-1000-09B
D-1401	XB11A-1000-19
D-1402	XB11A-1000-21
D-1403	XB11A-1000-21
D-1404	XB11A-1000-18
D-1405	XB11A-1000-18
D-1406	XB11A-1000-20
D-1407	XB11A-1000-20
D-1409	XB11A-1000-22
D-1501	XB11A-1000-23
D-1502	XB11A-1000-25
D-1503	XB11A-1000-24
D-1504	XB12A-1000-02

E-EXCHANGER

EQUIPMENT	DWG. NO.
E-1001A/B/C	XB11A-1000-05
E-1002	XB11A-1000-06
E-1003	XB11A-1000-07
E-1004	XB11A-1000-07
E-1005	XB11A-1000-07
E-1006	XB11A-1000-07
E-1015	XB12A-1000-07B
E-1101	XB11A-1000-12
E-1102A/B/C1/C2	XB11A-1000-12
E-1103	XB11A-1000-16
E-1104A/B	XB11A-1000-16
E-1105	XB11A-1000-13
E-1106A/B	XB11A-1000-13
E-1107	XB11A-1000-16
E-1108	XB11A-1000-15
E-1109	XB11A-1000-14
E-1110A/B	XB11A-1000-15
E-1201	XB11A-1000-10C
E-1202	XB11A-1000-10A
E-1203	XB11A-1000-10C
E-1301A	XB11A-1000-09A
E-1301B	XB11A-1000-09B
E-1302	XB11A-1000-09B
E-1404	XB11A-1000-21
E-1405	XB11A-1000-21
E-1501A/B	XB11A-1000-25
E-1502	XB11A-1000-24
E-1503A/B	XB11A-1000-23
E-1504	XB11A-1000-23
E-1505	XB12A-1000-02
E-1506	XB12A-1000-02

F-FURNACE

EQUIPMENT	DWG. NO.
F-1001	XB11A-1000-04A

M-MISCELLANEOUS

EQUIPMENT	DWG. NO.
M-1001	XB11A-1000-11
M-1002	XB12A-1000-01
M-1003	XB11A-1000-04B
M-1013	XB12A-1000-12A
M-1014	XB12A-1000-13A
M-1101A/B	XB11A-1000-17

P-PUMP

EQUIPMENT	DWG. NO.
P-1001A/B	XB11A-1000-02
P-1002A/B	XB12A-1000-03
P-1003A/B	XB12A-1000-04
P-1004A/B	XB12A-1000-04
P-1008A/B	XB12A-1000-04
P-1010A/B	XB12A-1000-10
P-1011A/B	XB12A-1000-09
P-1012A/S	XB12A-1000-11A
P-1013A/S	XB12A-1000-12A
P-1014	XB12A-1000-13A
P-1015A/B	XB12A-1000-07B
P-1101A/B	XB11A-1000-16
P-1102A/B	XB11A-1000-13
P-1103A/B	XB11A-1000-15
P-1201A/B	XB11A-1000-10C
P-1401A/B	XB11A-1000-21
P-1403A/B	XB11A-1000-20
P-1406A/B	XB11A-1000-22
P-1501A/B	XB11A-1000-25
P-1502A/B	XB11A-1000-24
P-1503A/B	XB11A-1000-23
P-1504A/B	XB11A-1000-25
P-1505A/B	XB12A-1000-02

R-REACTOR

EQUIPMENT	DWG. NO.
R-1001	XB11A-1000-05
R-1002	XB11A-1000-06

S-FILTER & STRAINER

EQUIPMENT	DWG. NO.
S-1001A/B/C	XB11A-1000-01B
S-1002A/B/C	XB11A-1000-01C
S-1003A/B	XB12A-1000-05
S-1004	XB11A-1000-01B
S-1005	XB11A-1000-01C
S-1405A/B	XB11A-1000-21

SL-SILENCER

EQUIPMENT	DWG. NO.
SL-1001	XB11A-1000-04B
SL-1002	XB12A-1000-01
SL-1003	XB12A-1000-03

T-TANK

EQUIPMENT	DWG. NO.
T-1002	XB12A-1000-04
T-1003	XB12A-1000-04
T-1004	XB12A-1000-04
T-1012	XB12A-1000-11A
T-1013	XB12A-1000-12A
T-1014	XB12A-1000-13A

V-TOWER

EQUIPMENT	DWG. NO.
V-1101	XB11A-1000-12
V-1102	XB11A-1000-14
V-1401	XB11A-1000-18
V-1402	XB11A-1000-20
V-1501	XB11A-1000-23

Att-2 (1/12)

PIPING SYMBOLS AND LEGENDS

- MAIN PROCESS LINES
- OTHER PROCESS AND UTILITY LINES
- PROCESS LINES WITH STEAM TRACING
- WATER SPARY
- J — J — J : PROCESS LINES WITH STEAM JACKETED
- GATE VALVE
- BALL VALVE
- DIAPHRAGM VALVE
- BUTTERFLY VALVE
- GLOBE VALVE
- CHECK VALVE
- STOP CHECK STRAIGHT VALVE
- STOP CHECK ANGLE VALVE
- PLUG VALVE
- THREE WAY PLUG VALVE
- FOUR WAY PLUG VALVE
- NEEDLE VALVE
- J.V. : JACKET VALVE
- SLIDE VALVE
- ARV : AUTOMATIC RECIRCULATION VALVE (ARV)
- PRESSURE SIDE / SEAT SIDE : ORBIT VALVE
- SPRING LOADED QUICK CLOSE VALVE FOR INTERMITTENT BLOWDOWN
- CONTINUOUS BLOWDOWN
- GATE VALVE WITH GLOBE BYPASS VALVE
- LINE CLASS CHANGE
- AA2U BA2U : ANGLE VALVE
- THREE-WAY VALVE
- FOUR-WAY VALVE
- EQUALIZING VALVE ON MAIN VALVE BODY
- WELD CAP
- BLIND FLANGE
- HOSE CONNECTION
- RS : REMOVABLE PIPE SPOOL
- SCREWED CAP
- FLANGE JOINT
- BURNER NOZZLE
- REDUCTION IN LINE SIZE
- PULSATION DAMPENER
- FREE DRAIN LINE (DO NOT POCKET LINE)
- SLOPE : SLOPED LINE
- FLEXIBLE HOSE
- EXPANSION JOINT

- CONTINUITY ARROW
- B/L : CONTINUITY ARROW (FROM BATTERY LIMIT)
- SPACER
- SPACER (NORMALLY CLOSED)
- SPECTACLE BLIND (NORMALLY OPEN)
- SPECTACLE BLIND (NORMALLY CLOSED)
- DAMPER
- OIL SEPARATOR
- SIGHT GLASS
- DRESSER COUPLING
- T : STEAM TRAP
- F : FILTER
- T B : STRAINER (B – BASKET, C – CONE, P – PLATE, T – TEMPORARY)
- BASKET TYPE STRAINER
- TEE TYPE STRAINER
- "Y" TYPE STRAINER
- STRAINER-DUAL
- VORTEX BREAKER
- MANOMETER
- VENT
- OPEN VENT
- LIQUID SEAL
- BREATHER VALVE
- FLAME ARRESTER
- SP : PIPING SPECIALTY ITEM
- S : SAMPLE CONNECTION WITHOUT COOLER
- SC : SAMPLE CONNECTION WITH COOLER
- TRAP-VACUUM BOOSTER (LIFT) CONTINUOUS DRAINER
- STEAM EXHAUST HEAD
- G.H. : GAS TIGHT GAUGE HATCH (WITHOUT VENTING GAS TO ATMOSPHERE)
- G.H. : GAUGE HATCH
- ABOVE GROUND / UNDER GROUND : DRAIN CONN. TO CLOSED SYSTEM
- AREA DRAIN FITTING
- OD : OILY DRAIN
- EQUIPMENT INSULATION
- PACKAGE EQUIPMENT OR INTEGRAL UNITS
- SUPPLIED BY VENDOR / SUPPLIED BY PURCHASER : SUPPLY LIMIT
- LINE CLASS BREAK
- INSULATION BREAK
- LINE NUMBER BREAK
- SPRAY NOZZLE DEVICE

LINE NOMENCLATURE

10" – P – 1 – 027 – JE2H – H – XX

- 10" : LINE SIZE
- P : FLUID SYMBOL
- 1 : UNIT NUMBER
- 027 : LINE NUMBER
- JE2H : MATERIAL CLASS
- H : INSULATION TYPE
- XX : INSULATION THICKNESS OR QUANTIT AND SIZE OF TRACING

Insulation type:
- B – ANTI-SWEAT
- C – COLD
- H – HOT
- P – PERSONAL PROTECTION
- T – STEAM TRACED
- A – ACOUSTIC
- E – ELECTRIC TRACER
- J – STEAM JACKETING

Unit number:

10~13	HDSU
14	AAU
15	SWTU

LINE CLASS DESIGNATION

1ST SYMBOL: RATING AND FACING

- A: ANSI 150 LB, RF
- B: ANSI 300 LB, RF
- C: ANSI 600 LB, RF
- D: ANSI 900 LB, RF
- E: ANSI 1500 LB, RF
- F: ANSI 2500 LB, RF
- G: ANSI 2500 LB, RJ
- H: ANSI 1500 LB, RJ
- I: ANSI 900 LB, RJ
- J: ANSI 600 LB, RJ
- K: ANSI 300 LB, RJ
- L: ANSI 150 LB, /125 LB, FF

2ND SYMBOL: PIPE MATERIAL

- A: CARBON STEEL
- B: KILLED CARBON STEEL
- C: CAST IRON
- D: CARBON STEEL, GALVANIZED
- E: 1¼ CR–½ MO STEEL
- F: 2¼ CR–1 MO STEEL
- G: 5 CR–½ MO STEEL
- H: 9 CR–1 MO STEEL
- J: 304 STAINLESS STEEL
- K: 316 STAINLESS STEEL
- L: 304L STAINLESS STEEL
- M: 316L STAINLESS STEEL
- N: 321 STAINLESS STEEL
- P: 347 STAINLESS STEEL
- R: CARBON STEEL (LOW TEMP. SERVICE)
- S: 3½ NI STEEL
- T: C.S. WITH PTFE LINING
- U: 304H STAINLESS STEEL

3RD SYMBOL: CORROSION ALLOWANCE

- 0: 0" (0 MM)
- 1: 0.039" (1.0 MM)
- 2: 0.063" (1.6 MM)
- 3: 0.100" (2.5 MM)
- 4: 0.125" (3.0 MM)
- 5: 0.188" (4.8 MM)
- 6: 0.250" (6.3 MM)

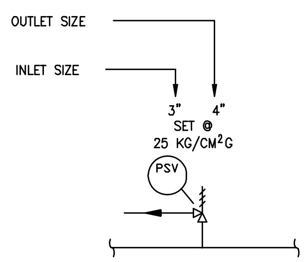
4TH SYMBOL: SPECIAL SERVICE

- A: AMINE
- B: ANSI B.31.1 PIPING
- C: HYDROGEN AND SOUR SERVICE
- H: HYDROGEN
- J: JACKETING
- S: NACE SOUR SERVICE
- U: UTILITY (STEAM, AIR,NITROGEN, AND CONDENSATE ETC.)
- W: FIREWATER, INDUSTRICAL WATER, COOLING WATER
- X: JACKETING AND SOUR SERVICE

VALVE POSITION

CSC	CAR SEAL CLOSED
CSO	CAR SEAL OPEN
FC	FAIL CLOSED
FL	FAILURE LOCK
FLC	FAILURE LOCK TO CLOSE
FO	FAIL OPEN
LC	LOCKED CLOSED
LO	LOCKED OPEN
NC	NORMALLY CLOSED (OR)
NO	NORMALLY OPEN
TSO	TIGHT SHUT-OFF(NOTE 5)

LEGEND FOR SAFETY VALVE



(F) FURNISHED WITH EQUIPMENT

FLUID SYMBOL FOR LINE SERVICE

SERVICE SYSTEM	FLUID	SYMBOL
AIR SYSTEM	INSTRUMENT AIR	IA
	PLANT AIR	PA
	DRY AIR	DA
BLOWDOWN AND PUMPOUT	INTERMITTENT BLOWDOWN	BB
	CONTINUOUS BLOWDOWN	CB
	SLOP	SB
	BLOWDOWN	BD
DRAIN	CHEMICAL OR CONTAMINATED DRAIN	CD
	AMINE DRAIN	ED
	LIQUID DRAIN	LD
	OIL DRAIN	OD
	OILY SEWER	OS
	NON-CONTAMINATED DRAIN	ND
	SOUR WATER DRAIN	WD
	ACID DRAIN	AD
	WATER SEWER	WS
	SPENT CAUSTIC DRAIN	CLD
FLARE SYSTEM	H_2S FLARE	XF
	HOT FLARE	HF
	WET FLARE	WF
	DRY FLARE	DF
GAS SYSTEM	FUEL GAS	FG
	NITROGEN	NG
	NATURAL GAS	GG
OIL UTILITY SYSTEM	FLUSH OIL	FO
	SEAL OIL	SO
	FLUSHING OIL PURGE OIL	WO
	HYDRULIC OIL	HO
	INJECTION OIL	IO
	LUBE OIL	LO
	QUENCH OIL	QO
	FLUX OIL	XO
	DISPLACEMENT NAPHTHA (ATMOSPHERIC NAPHTHA)	DN
	DISPLACEMENT OIL (ATMOSPHERIC GAS OIL)	DO
PROCESS	PROCESS SERVICE	P
STEAM AND CONDENSATE	HIGH PRESSURE CONDENSATE	HC
	MEDIUM PRESSURE CONDENSATE	MC
	LOW PRESSURE CONDENSATE	LC
	HIGH PRESSURE STEAM	HS
	HIGH PRESSURE SUPERHEATED STEAM	XS
	MEDIUM PRESSURE STEAM	MS
	MEDIUM PRESSURE SUPERHEATED STEAM	TS
	LOW PRESSURE STEAM	LS
	LOW PRESSURE SUPERHEATED STEAM	SS
	EXHAUSTED STEAM	ES
	DILUTION STEAM	DS
VENT	ATMOSPHERE	AV
	SAFETY VALVE TO ATMOSPHERE	SV
WATER SYSTEM	COOLING WATER SUPPLY	SW
	COOLING WATER RETURN	RW
	TEMPERED WATER SUPPLY	LW
	INTERMEDIATE COOLING WATER	MW
	TEMPERED WATER RETURN	LW
	FIRE WATER	FW
	DRINK WATER	DW
	UTILITY WATER (INDUSTRIAL WATER)	UW
	TREATED WATER (DEMINERALIZED WATER)	TW
	DEAERATED BOILER FEED WATER	BW
	WASH WATER (FLUSH WATER & PURGE WATER)	XW
	PROCESS WATER	PW
	WASTE WATER	WW
	QUENCH WATER	QW
SPECIAL LIQUIDS	CAUSTIC	CL
	CHEMICAL INJECTION	IL
	ACID	AL
	MEA, DIPA, DEA	EL
	SOLVENT	SL
	SULFOLANE	SU
	PDEB	DE

GENERAL NOTES:

1. THE DRAIN VALVE UPSTREAM THE CONTROL VALVE SHALL BE ¾"
2. PUMP CASING VENT AND DRAIN PIPE SHALL BE EXTENDED TO OILY SEWER.
3. HIGH POINTS OF ALL LINES SHALL BE VENTED AND LOW POINTS DRAINED EXCEPT WHERE THE LINE CAN BE VENTED OR DRAINED ELSE WHERE AND GENERALLY. ABOVE VENTS END DRAINS ARE NOT NOTED ON THE P & I DIAGRAM.
4. A BLEED VALVE FOR A PROCESS BLOCK VALVE SHALL BE LOCATED AS CLOSE AS POSSIBLE TO THE BLOCK VALVE.
5. THE LEAKAGE RATE OF TIGHTLY SHUT-OFF CONTROL VALVE SHALL BE BETTER THAN ANSI CLASS V.

Att-2 (2/12)

INSTRUMENT IDENTIFICATION LETTERS

FIRST-LETTERS	INITIATING OR MEASURED VARIABLE	PRIMARY ELEMENT	TRANSMITTERS	WELL OR PROBE	READOUT DEVICES		CONTROLLERS		CONTROL VALVES	SWITCH	ALARM	VIEWING DEVICE GLASS	SAFETY DEVICE	SOLENOIDE. RELAYS. COMPUTING DEVICES
					INDICATING	RECORDING	DCS CONTROLLER	LOCAL CONTROLLER						
A	ANALYSIS	AE	AT		AI	AR	AC	AIC		AS	AA			AY
B	BURNER FLAME	BE								BS	BA	BG		
F	FLOW RATE	FE	FT		FI	FR	FC	FIC	FV	FS	FA	FG		FY
H	HAND						HC	HIC	HV	HS				HY
L	LEVEL		LT		LI	LR	LC	LIC	LV	LS	LA	LG		LY
P	PRESSURE/VACUUM		PT		PI	PR	PC	PIC	PV	PS	PA	PG	PSV	PY
PD	PRESSURE.DIFFERENTIAL		PDT		PDI		PDC					PDG		PDY
T	TEMPERATURE	TE	TT	TW		TR	TC	TIC	TV	TS	TA	TG		TY
V	VIBRATION									VS	VA			
Z	POSITION/DIMENSION				ZI					ZS				
	OTHER	_E	_T	_W	_I	_R	_C	_IC	_V	_S	_A	_G		_Y

* ADD SUFFIX LETTERS H. L. HH. OR LL TO SWITCHES AND ALARMS TO INDICATE HIGH. LOW. HIGH HIGH. OR LOW LOW.

PREFIX LETTERS FOR OTHER VARIABLES

C	CONDUCTIVITY
D	DENSITY
E	VOLTAGE
FF	FLOW RATIO
I	CURRENT/ELECTRIC CURRENT
J	POWER
K	TIME
M	MOISTURE/HUMIDITY
Q	QUANTITY OR EVENT
R	RADIOACTIVITY
S	SPEED OR FREQUENCY
W	WEIGHT/FORCE
X	RUNNING

INSTRUMENT NOMENCLATURE

FIC – 1 0 0 1

SEQUENCE NUMBER

UNITNUMBER

10~13	HDS
14	AAU
15	SWTU

INSTRUMENT FUNCTION

OTHER ABBREVIATIONS

B.L.	BATTERY LIMIT
CC	CHEMICAL CONNECTION
C.W.	COOLING WATER
EBD	EMERGENCY BLOWDOWN
ESD	EMERGEMCY SHUTDOWN
HH	HANDHOLE
H_2	HYDROGEN ANALYZER
H_2S	HYDROGEN SULFIDE ANALYZER
IA	AIR SUPPLY
JV	JACKET VALVE
MH	MANHOLE
NNF	NORMALLY NO FLOW
O_2	OXYGEN ANALYZER
P	POINT (TEST CONN.)
P_H	P_H METER
PB	PUSH BUTTON SWITCH
RO	RESTRICTION ORIFICE
RTD	RESISTANCE TEMP ELEMENT
S/D	SHUTDOWN
SI	SPEED INDICATOR
SO	STEAM OUT
SP	SET POINT
SR	SPLIT RANGE
UA	COMMON PREALARM
UC	UTILITY CONNECTION
VB	VACUUM BREAKER
VISC.	VISCOSITY
XA	EQUIPMENT S/D ALARM OR MALFUNCTION
XI	RUNNING STATUS
XIC	SPEED CONTROL
PA	PURGE FAILURE

RELAY FUNCTION DESIGNATIONS

TYPES

Σ	=	ADD
Δ	=	SUBTRACT
±	=	BIAS
÷	=	DIVIDE
×	=	MULTIPLY
√	=	EXTRACT SQUARE ROOT
>	=	HIGH SELECT
<	=	LOW SELECT
K	=	CONSTANT FACTOR
‖	=	BOOSTER
REV.	=	REVERSE
P/I	=	PULSE TO CURRENT
I/P	=	CURRENT TO PULSE
I/P	=	CURRENT TO PNEUMATIC
P/I	=	PNEUMATIC TO CURRENT
E/I	=	VOLTAGE TO CURRENT

FLUSHING (O – OIL) (W – WATER)

PURGE (W – WATER) (H2 – HYDROGEN) (O – OIL)

R — RESET FOR LATCH TYPE ACTUATOR

INTERLOCK IF ALL INPUTS EXIST

OR — INTERLOCK IF ANY ONE OR MORE INPUTS EXIST

INTERLOCK LOGIC

INSTRUMENT IDENTIFICATION SYMBOLS

LOCALLY MOUNTED INSTRUMENT

BOARD MOUNTED INSTRUMENT

INSTRUMENT FURNISHED WITH PROCESS EQUIPMENT

INSTRUMENT MOUNTED BEHIND BOARD OR ON RACK

CP — LOCAL PANEL MOUNTED INSTRUMENT (CP – COMPRESSOR PANEL. HP – HEATER PANEL) (LP – LOCAL PANEL IN LOCAL CONTROL ROOM)

DISTRIBUTED CONTROL SYSTEM. BLIND CONTROLLER/COMPUTATION/SIGNAL. CONDITIONING OR SHARED DISPLAY INSTALLED IN FIELD.NOT NORMALLY ACCESSIBLE TO OPERATOR

DISTRIBUTED CONTROL SYSTEM. INDICATOR/CONTROLLER/RECORDER OR ALARM POINT. NORMALLU ACCESSIBLE TO OPERATOR.

MAL — MALFUNCTION ALARM

HS C–S–O — MOTOR OPERATED VALVE SWITCH ON DCS WITH CLOSE–STOP–OPEN FUNCTION

HS C–S–O — MOTOR OPERATED VALVE SWITCH ON BOARD WITH CLOSE–STOP–OPEN FUNCTION

R — DISTRIBUTED CONTROL SYSTEM WITH TREND MEMORY CAPABILITY

H L — DISTRIBUTED CONTROL SYSTEM WITH SOFT–WIRED HIGH (LOW) ALARM FUNCTION

HH LL — DISTRIBUTED CONTROL SYSTEM WITH SOFT–WIRED HIGH HIGH (LOW LOW) ALARM FUNCTION

HEAT TRACED INSTRUMENT

FT XXX — IN–LINE TYPE FLOW INSTRUMENT PDM (DISPLACEMENT METER) MAG (MAGNETIC FLOWMETER) INTEG.–(INTEGRAL ORIFICE) ULTRA SONIC

PILOT LIGHT (R–RED. G–GREEN. A–AMBER)

PROCESS COMPUTER NOT NORMALLY ACCESSIBLE TO OPERATOR

PROCESS COMPUTER NORMALLY ACCESSIBLE TO OPERATOR

PACKAGED PROGRAMMABLE LOGIC CONTROLLER OR DIGITAL LOGIC CONTROL INTEGRAL TO THE DISTRIBUTED CONTROL SYSTEM. NOT NORMALLY ACCESSIBLLE TO OPERATOR

PACKAGED PROGRAMMABLE LOGIC CONTROLLER OR DIGITAL LOGIC CONTROL INTEGRAL TO THE DISTRIBUTED CONTROL SYSTEM. NORMALLY ACCESSIBLE TO OPERATOR

ELECTRIC SWITCH CHANGE–OVER

INSTRUMENT WITH TWO FUNCTIONS

INSTRUMENT SUPPLY OR CONNECTION TO PROCESS

INSTRUMENT LINE. PNEUMATIC SIGNAL

INSTRUMENT LINE. ELECTRICAL SIGNAL

X X — INSTRUMENT LINE. CAPILLARY TUBE

L L — INSTRUMENT LINE. HYDRAULIC SIGNAL

E E — INSTRUMENT LINE. ELECTRIC TRACING

INSTRUMENT LINE. SOFTWARE OR COMMUNICATION LINK

INSTRUMENT LINERADIOACTIVE. SONIC OR LIGHT SIGNAL

CONTROL VALVE

CONTROL VALVE WITH MANUAL OPERATION

CONTROL DAMPER

HAND OPERATED CONTROL VALVE

M — MAGNETIC FLOW METER

PRESSURE CONTROL VALVE. EXTERNAL TAP

PRESSURE CONTROL VALVE. SELF CONTAINED

FLOAT OPERATED VALVE

AIR CYLINDER OPERATED VALVE

H — ELECTRO HYDRAULIC OPERATED VALVE

MOV — ELECTRO MOTOR OPERATED VALVE

S — SOLENOID VALVE

S FO — THREE–WAY SOLENOID VALVE LATCH TYPE WITH MANUAL RESET. SHOWING DEENERGIZED POSITION

S FO — THREE–WAY SOLENOID VALVE. SHOWING DEENERGIZED POSITION

FOUR WAY SOLENOID VALVE

DOUBLE DIAPHRAGM CONTROL VALVE

DIAPHRAGM OPERATED BUTTERFLY VALVE

DIAPHRAGM OPERATED BALL VALVE OR ECCNTRIC DISC VALVE

DIAPHRAGM OPERATED ANGLE VALVE

RELIEF VALVE

PRESSURE OR VACUUM RELIEF VALVE

RUPTURE DISC FOR PRESSURE RELIEF

RUPTURE DISC FOR VACUUM RELIEF

DIAPHRAGM SEAL

CHEMICAL SEAL

ORIFICE PLATE

VORTEX FLOW ELEMENT

POSITIVE DISPLACEMENT METER

TURBINE OR PROPELLER TYPE FLOW ELEMENT

VENTURI

ANNUBAR

FLOW NOZZLE

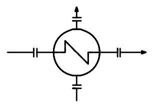
FI — ROTAMETER

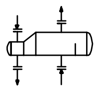
FI — ROTAMETER WITH INTEGRAL THROTTLE VALVE

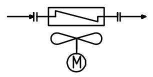
MASS — MASS FLOW METER

Att-2 (3/12)

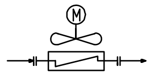
EQUIPMENT SYMBOLS


SHELL & TUBE EXCHANGER

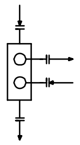

KETTLE TYPE REBOILER

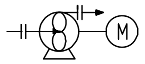


FORCED DRAFT TYPE AIR COOLER

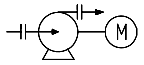


INDUCED DRAFT TYPE AIR COOLER

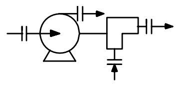

DOUBLE PIPE EXCHANGER

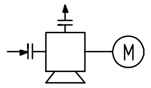


MOTOR DRIVEN ROTARY PUMP

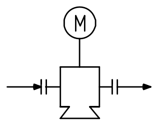


MOTOR DRIVEN CENTRIFUGAL PUMP

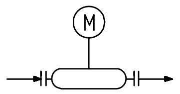

TURBINE DRIVEN CENTRIFUGAL PUMP



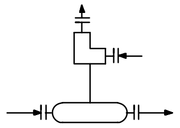
MOTOR DRIVEN RECIPROCATING PUMP

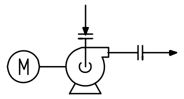


MOTOR DRIVEN SUMP OR
SUBMERGED PUMP
(VERTICAL CENTRI.)

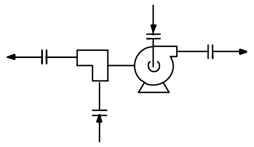


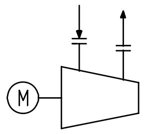
MOTOR DRIVEN IN LINE PUMP


TURBINE DRIVEN IN LINE PUMP

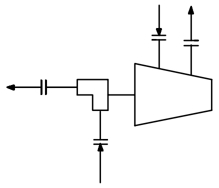


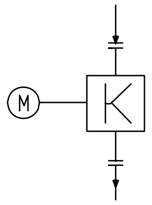
MOTOR DRIVEN FAN OR BLOWER


TURBINE DRIVEN FAN OR BLOWER

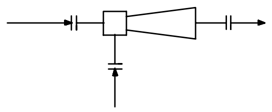


MOTOR DRIVEN CENTRIFUGAL
COMPRESSOR

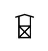

TURBINE DRIVEN CENTRIFUGAL
COMPRESSOR

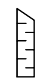


MOTOR DRIVEN RECIPROCATING
COMPRESSOR


EJECTOR


FILTER–AIR INTAKE

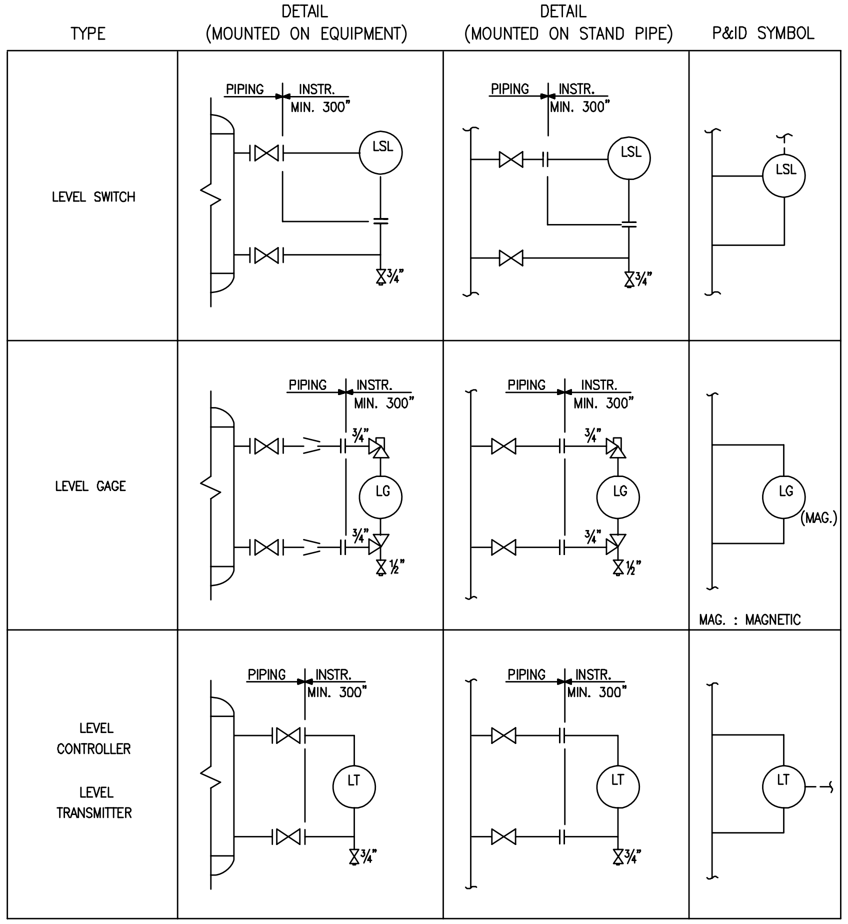

BLOWER INLET FILTER.
WITH SILENCER


SILENCER

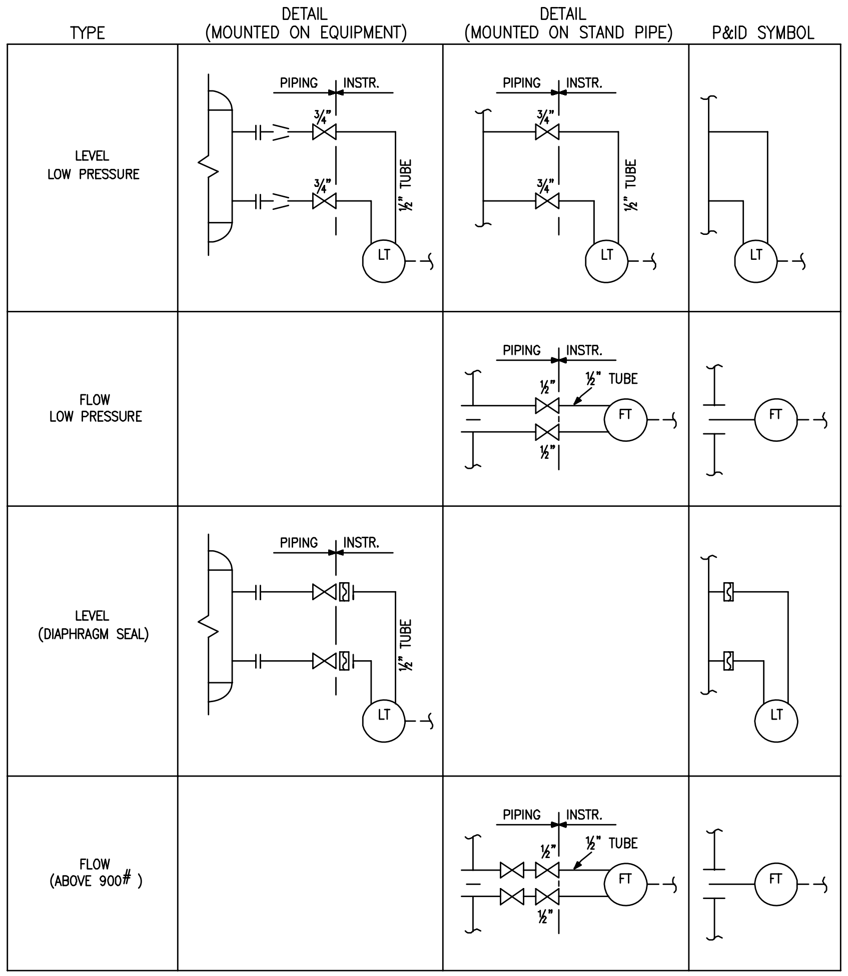
Att-2 (4/12)

LEVEL INSTRUMENTS

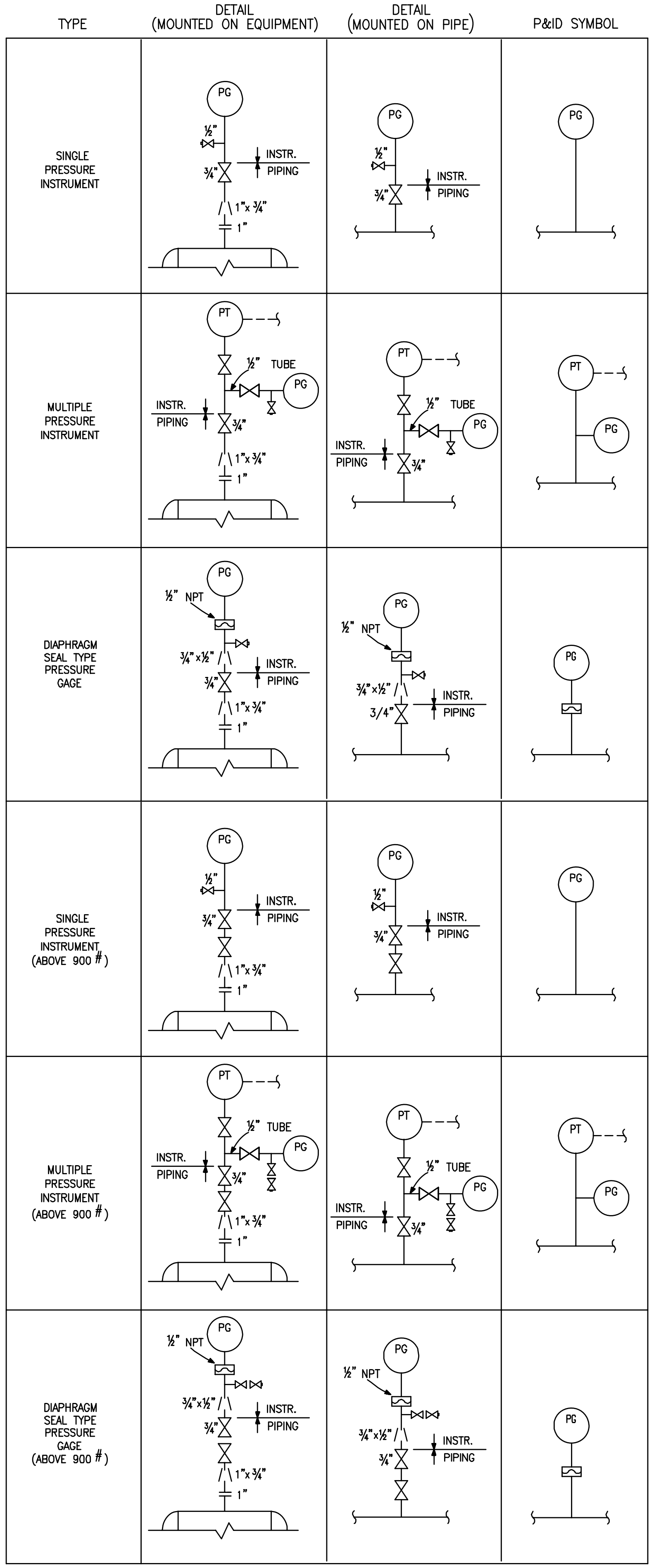
LEVEL SWITCHES.GAGE GLASSES.
(EXTERNAL FLOAT BUT NOT DP CELLS)



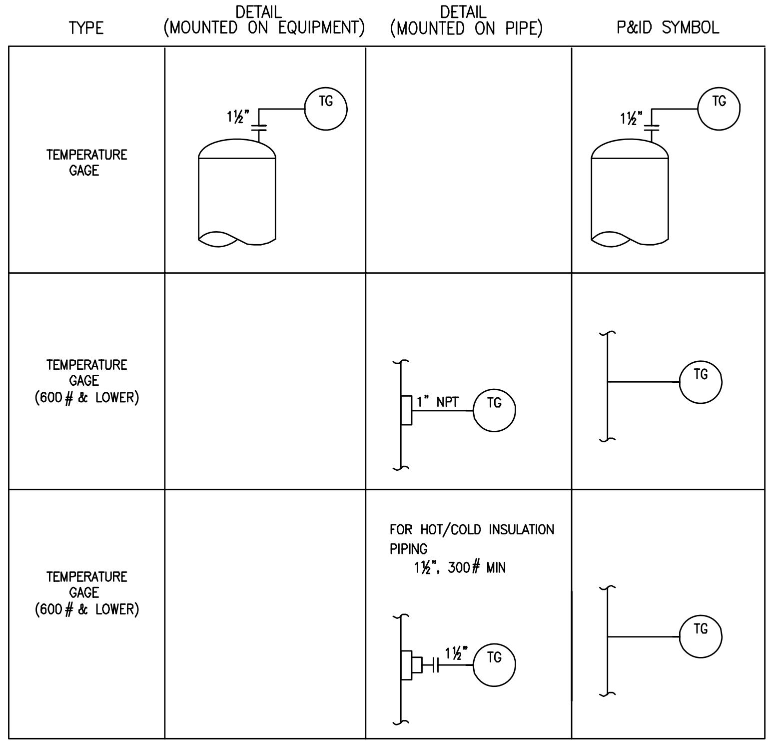
DP INSTRUMENT

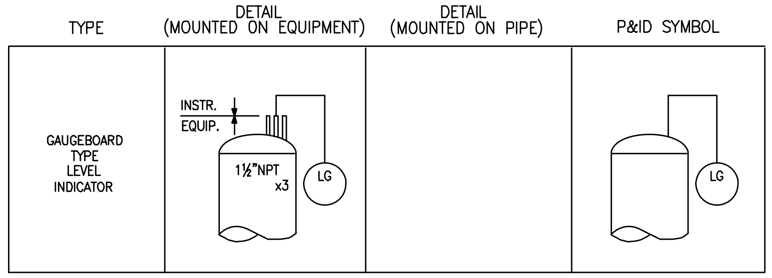


PRESSURE INSTRUMENTS



TEMPERATURE INSTRUMENTS (TE/TW/TG)

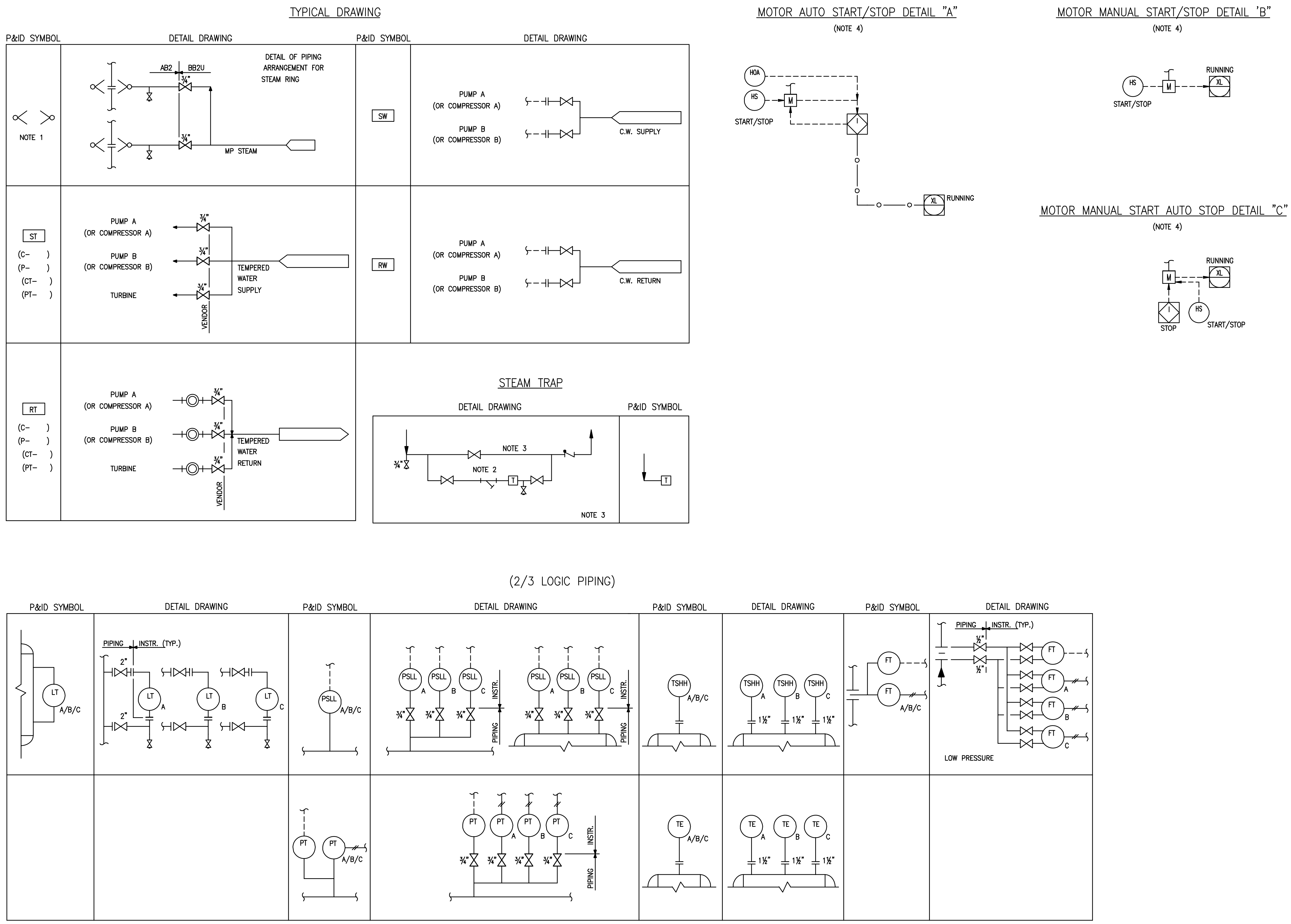




NOTES:

1. FOR TYPICAL BRIDLE (MOUNTED) DP CELL. DISPLACEMENT LT. LG & LS. PLEASE SEE DWG.XB11A–0000–07 P&ID SYMBOL IS THE SAME AS THIS DWG.
2. INSTRUMENT CONNECTION ON JACKETED LINE SHALL BE REFERED TO EACH P&ID.

Att-2 (5/12)

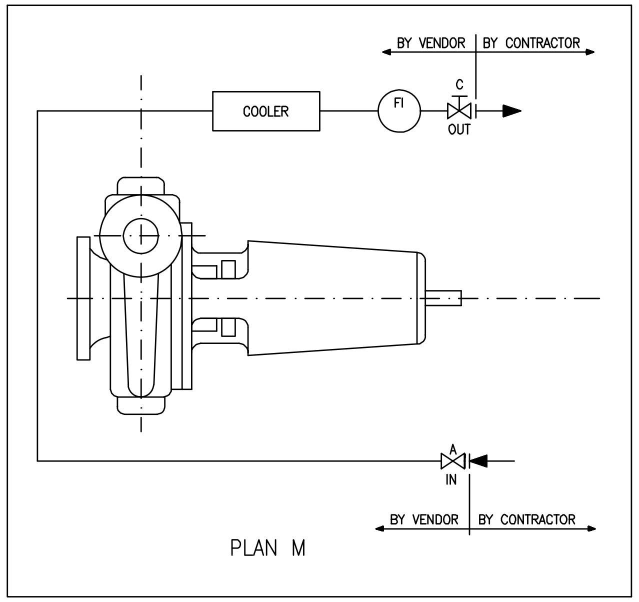


NOTES:

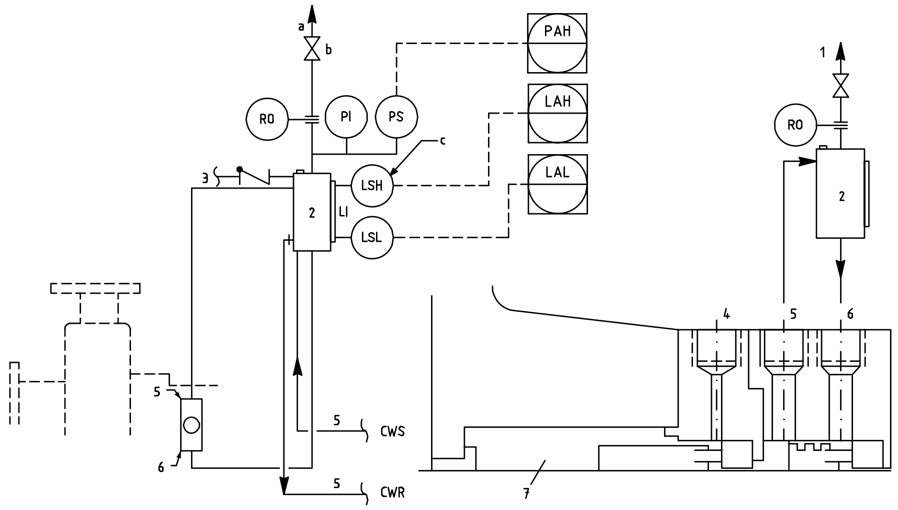
1. VALVES SHOULD BE AT GRADE.
2. TRAP WITH INTEGRAL STRAINER. THE Y-STRAINER CAN BE DELETED.
3. THE BYPASS LINE AND VALVE CAN BE DELETED FOR LOW POINT TRAP SETS.
4. ALL INSTRUMENT TAG NO. (STATUS & SWITCHES) WILL BE THE SAME AS MOTOR (EQUIPMENT) TAG NO. THE DETAILS WILL BE APPLICABLE TO ALL ISBL & OSBL P&IDS EXCEPT FOR FIRE WATER SYSTEM. THE ADDITIONAL FUNCTIONS SHOULD BE INDICATED ON RELEVANT P&ID.

Att-2 (6/12)

PUMP API COOLING PLAN



PLAN M



KEY

- 1 TO COLLECTION SYSTEM
- 2 RESERVOIR
- 3 MAKE-UP BUFFER FLUID
- 4 FLUSH (F)
- 5 LIQUID BUFFER OUTLET (LBO)
- 6 LIQUID BUFFER INLET (LBI)
- 7 SEAL CHAMBER

- LSH = LEVEL SWITCH HIGH
- LSL = LEVEL SWITCH LOW
- LI = LEVEL INDICATOR

- PI = PRESSURE INDICATOR
- PSH = PRESSURE SWITCH HIGH

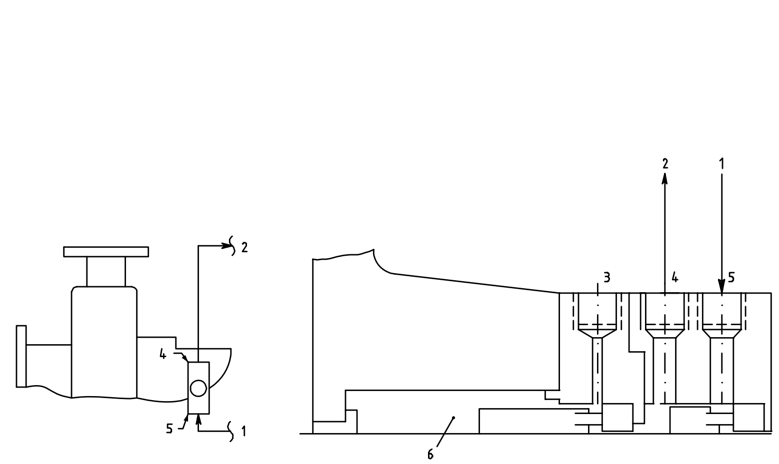
EXTERNAL RESERVOIR PROVIDING BUFFER FLUID FOR THE OUTER SEAL OF AN ARRANGEMENT 2 SEAL. DURING NORMAL OPERATION, CIRCULATION IS MAINTAINED BY AN INTERNAL PUMPING RING. THE RESERVOIR IS USUALLY CONTINUOUSLY VENTED TO A VAPOR RECOVERY SYSTEM AND IS MAINTAINED AT A PRESSURE LESS THAN THE PRESSURE IN THE SEAL CHAMBER.

a. ITEMS ABOVE THIS LINE ARE THE RESPONSIBILITY OF THE PURCHASER;
ITEMS BELOW THIS LINE SHALL BE SUPPLIED BY THE VENDOR.

b. NORMALLY OPEN.

c. IF SPECIFIED.

API PLAN 52



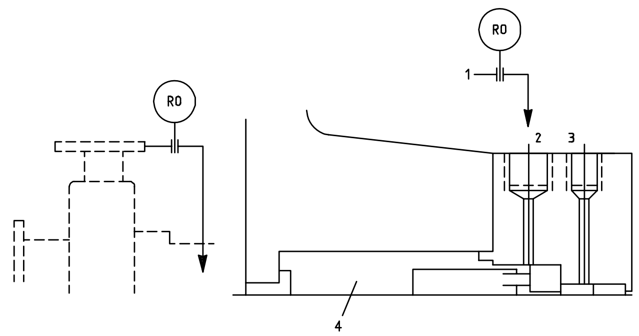
KEY

- 1 FROM EXTERNAL SOURCE
- 2 TO EXTERNAL SOURCE
- 3 FLUSH (F)
- 4 LIQUID BARRIER OUTLET (LBO)
- 5 LIQUID BARRIER INLET (LBI)
- 6 SEAL CHAMBER

RESSURIZED EXTERNAL BARRIER FLUID RESERVOIR OR SYSTEM SUPPLYING CLEAN FLUID TO THE SEAL CHAMBER. CIRCULATION IS BY AN EXTERNAL PUMP OR PRESSURE SYSTEM. RESERVOIR PRESSURE IS GREATER THAN THE PROCESS PRESSURE BEING SEALED. THIS PLAN IS USED WITH AN ARRANGEMENT 3 SEAL.

API PLAN 54

MECHANICAL SEAL PIPING

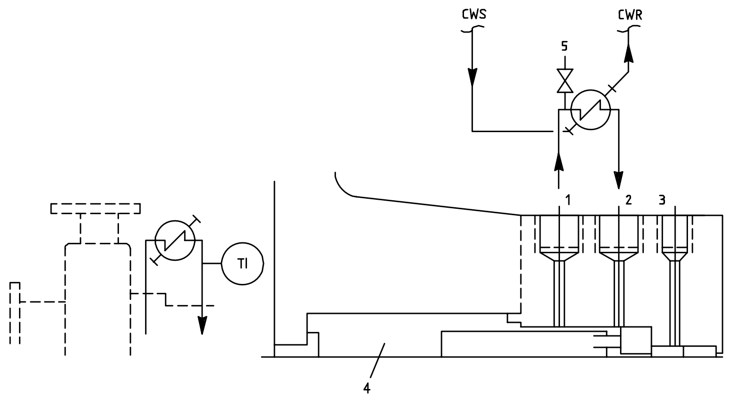


KEY

- 1 FROM PUMP DISCHARGE
- 2 FLUSH (F)
- 3 QUENCH/DRAIN (Q/D)
- 4 SEAL CHAMBER

RECIRCULATION FROM PUMP DISCHARGE THROUGH A FLOW CONTROL ORIFICE TO THE SEAL. THE FLOW ENTERS THE SEAL CHAMBER ADJACENT TO THE MECHANICAL SEAL FACES, FLUSHES THE FACES, AND FLOWS ACROSS THE SEAL BACK INTO THE PUMP.

API PLAN 11

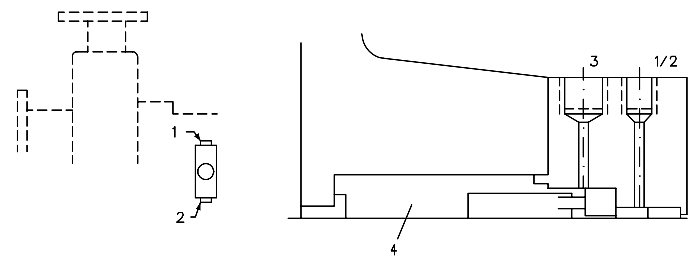


KEY

- 1 FLUSH OUTLET (FO)
- 2 FLUSH INLET (FI)
- 3 QUENCH/DRAIN, PLUGGED (Q/D)
- 4 SEAL CHAMBER
- 5 VENT
- TI TEMPERATURE INDICATOR

RECIRCULATION FROM A PUMPING RING IN THE SEAL CHAMBER THROUGH A COOLER AND BACK INTO THE SEAL CHAMBER. THIS PLAN CAN BE USED ON HOT APPLICATIONS TO MINIMIZE THE HEAT LOAD ON THE COOLER BY COOLING ONLY THE SMALL AMOUNT OF LIQUID THAT IS RECIRCULATED.

API PLAN 23



KEY

- 1 QUENCH (PLUGGED)
- 2 DRAIN (PLUGGED)
- 3 FLUSH
- 4 SEAL CHAMBER

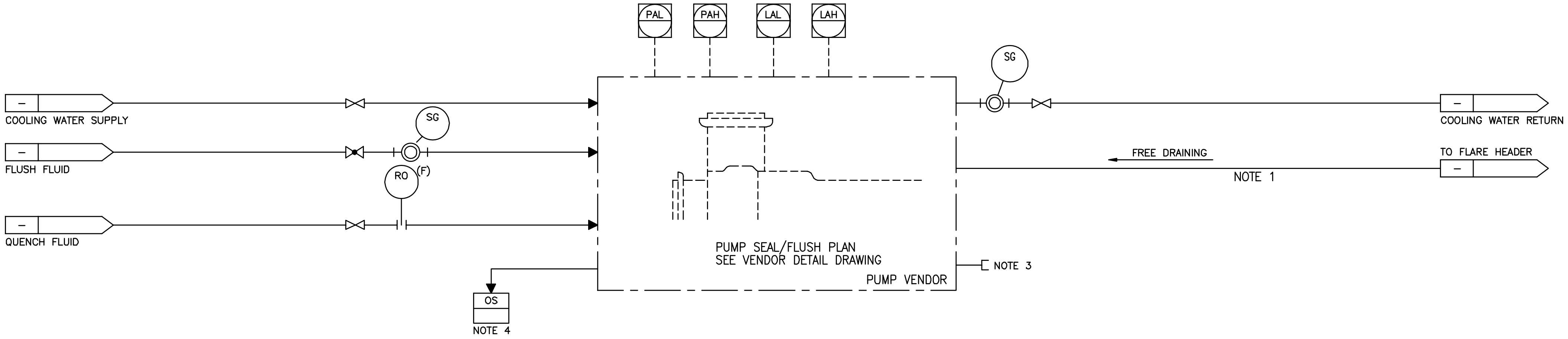
TAPPED AND PLUGGED CONNECTIONS FOR THE PURCHASER'S USE.
TYPICALLY THIS PLAN IS USED WHEN THE PURCHASER IS TO PROVIDE FLUID (SUCH AS STEAM, GAS, OR WATER) TO AN EXTERNAL SEALING DEVICE.

API PLAN 61

Att-2 (7/12)

NOTES:

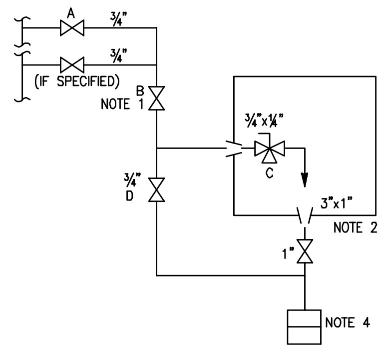
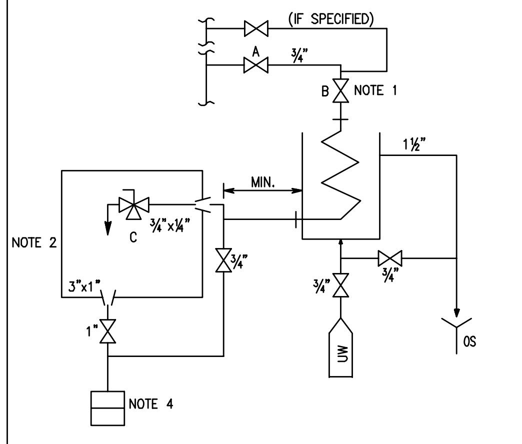
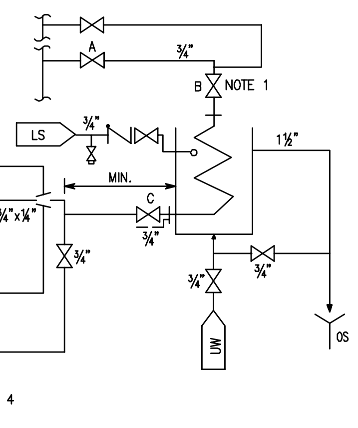
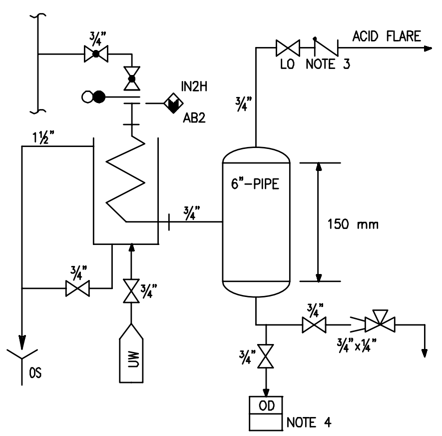
1. THE REQUIREMENT OF HEAT TRACING SHALL BE SPECIFIED ON THE LINE NO.
2. 各 EXTERNAL SUPPLY FLUID 及 LINE NO. 僅爲初步資料, FINALIZED 儀控及配管設計應依廠商資料爲準但 LINESIZE 應維持不小於設備之 NOZZLE SIZE.
3. DELETED.
4. FOR PLAN 62 QUENCH WATER DRAIN.



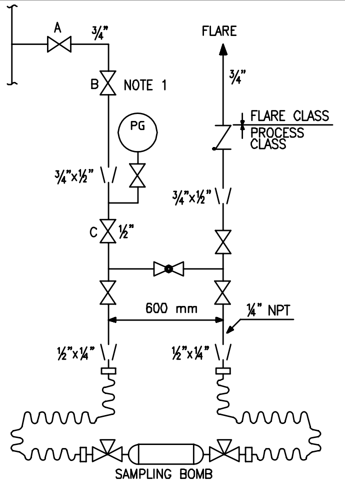
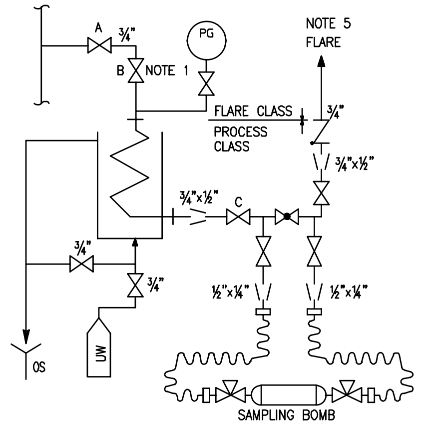
PUMP NO.	P&ID NO.	API PLAN	C.W. SUPPLY LINE NO.	C.W. RETURN LINE NO.	FLUSH/QUENCH FLUID LINE NO.	OILY DRAIN LINE NO.	RO. TAG NO.	SG TAG NO.	FLARE LINE NO.	DCS FUNCTION TAG NO.				REMARKS
										PAL	PAH	LAL	LAH	
P-1001 A/B	XB11A-1000-02	11+52	1½" SW 1601A/B-AB2W	1½" RW 1601A/B-AB2W	–	–	–	P1001A/B	¾" WF 1601A/B-AB2	–	P1001A1/A2/ B1/B2	P1001A1/A2/ B1/B2	P1001A1/A2/ B1/B2	
P-1002 A/B	XB12A-1000-03	23+61	1½" LW 1601A/B-AB2U	1½" LW 1602A/B-AB2U	–	–	–	P1002A/B	–	–	–	–	–	
P-1010 A/B	XB12A-1000-10	11+52	1" SW 1602A/B-AB2W	1" RW 1602A/B-AB2W	–	–	–	P1010A/B	¾" WF 1602A/B-AB2	–	P1010A/B	P1010A/B	P1010A/B	
P-1011 A/B	XB12A-1000-09	11+52	1" SW 1603A/B-AB2W	1" RW 1603A/B-AB2W	–	–	–	P1011A/B	¾" XF 1601A/B-AB2	–	P1011A/B	P1011A/B	P1011A/B	
P-1012 A/S	XB12A-1000-11A	54	1" LW 1607-AB2U	1" LW 1608-AB2U	–	–	–	P1012A	–	–	–	–	–	
P-1013 A/S	XB12A-1000-12A	54	1" LW 1609-AB2U	1" LW 1610-AB2U	–	–	–	P1013A	–	–	–	–	–	
P-1014	XB12A-1000-13A	54	1" LW 1611-AB2U	1" LW 1612-AB2U	–	–	–	P1014	–	–	–	–	–	
P-1015 A/B	XB12A-1000-07B	11	–	–	–	–	–	–	–	–	–	–	–	
P-1101 A/B	XB11A-1000-16	23+52	1" LW 1603A/B-AB2U	1" LW 1604A/B-AB2U	–	–	–	P1101A/B	¾" WF 1603A/B-AB2	–	P1101A/B	P1101A/B	P1101A/B	
P-1102 A	XB11A-1000-13	11+52	1" LW 1605A-AB2U	1" LW 1606A-AB2U	–	–	–	P1102A	¾" XF 1602A-AB2	–	P1102A	P1102A	P1102A	
P-1102 B	XB11A-1000-13	11+52	1" LW 1605B-AB2U	1" LW 1606B-AB2U	–	–	–	P1102B	¾" XF 1602B-AB2	–	P1102B	P1102B	P1102B	
P-1103 A/B	XB11A-1000-15	11+52	1" SW 1610A/B-AB2W	1" RW 1610A/B-AB2W	–	–	–	P1103A/B	¾" WF 1607A/B-AB2	–	P1103A/B	P1103A/B	P1103A/B	
P-1201 A/B	XB11A-1000-10C	11	–	–	–	–	–	–	–	–	–	–	–	
P-1401 A/B	XB11A-1000-21	11+52	1" SW 1604A/B-AB2W	1" RW 1604A/B-AB2W	–	–	–	P1401A/B	¾" WF 1606A/B-AB2	–	P1401A1/A2/ B1/B2	P1401A1/A2/ B1/B2	P1401A1/A2/ B1/B2	
P-1403 A/B	XB11A-1000-20	11+52	1" SW 1605A/B-AB2W	1" RW 1605A/B-AB2W	–	–	–	P1403A/B	¾" WF 1604A/B-AB2	–	P1403A/B	P1403A/B	P1403A/B	
P-1501 A/B	XB11A-1000-23	11+52	1" SW 1606A/B-AB2W	1" RW 1606A/B-AB2W	–	–	–	P1501A/B	¾" XF 1603A/B-AB2	–	P1501A/B	P1501A/B	P1501A/B	
P-1502 A/B	XB11A-1000-24	23+52	1" SW 1607A/B-AB2W	1" RW 1607A/B-AB2W	–	–	–	P1502A/B	¾" XF 1604A/B-AB2	–	P1502A/B	P1502A/B	P1502A/B	
P-1503 A/B	XB11A-1000-23	23+52	1" SW 1608A/B-AB2W	1" RW 1608A/B-AB2W	–	–	–	P1503A/B	¾" WF 1605A/B-AB2	–	P1503A/B	P1503A/B	P1503A/B	
P-1504 A/B	XB11A-1000-25	–	–	–	–	–	–	–	–	–	–	–	–	
P-1505 A/B	XB12A-1000-02	23+61	1" SW 1609A/B-AB2W	1" RW 1609A/B-AB2W	–	–	–	P1505A/B	–	–	–	–	–	
PT-1015A	XB12A-1000-07B	–	1" LW 1613-AB2U	1" LW 1614-AB2U	–	–	–	–	–	–	–	–	–	
PT-1101B	XB12A-1000-16	–	1" LW 1603T-AB2U	1" LW 1604T-AB2U	–	–	–	–	–	–	–	–	–	
PT-1102B	XB12A-1000-13	–	1" LW 1605T-AB2U	1" LW 1606T-AB2U	–	–	–	–	–	–	–	–	–	
PT-1201B	XB12A-1000-10C	–	1" LW 1042-AB2U	1" LW 1043-AB2U	–	–	–	–	–	–	–	–	–	

Att-2 (8/12)

SAMPLE STATIONS

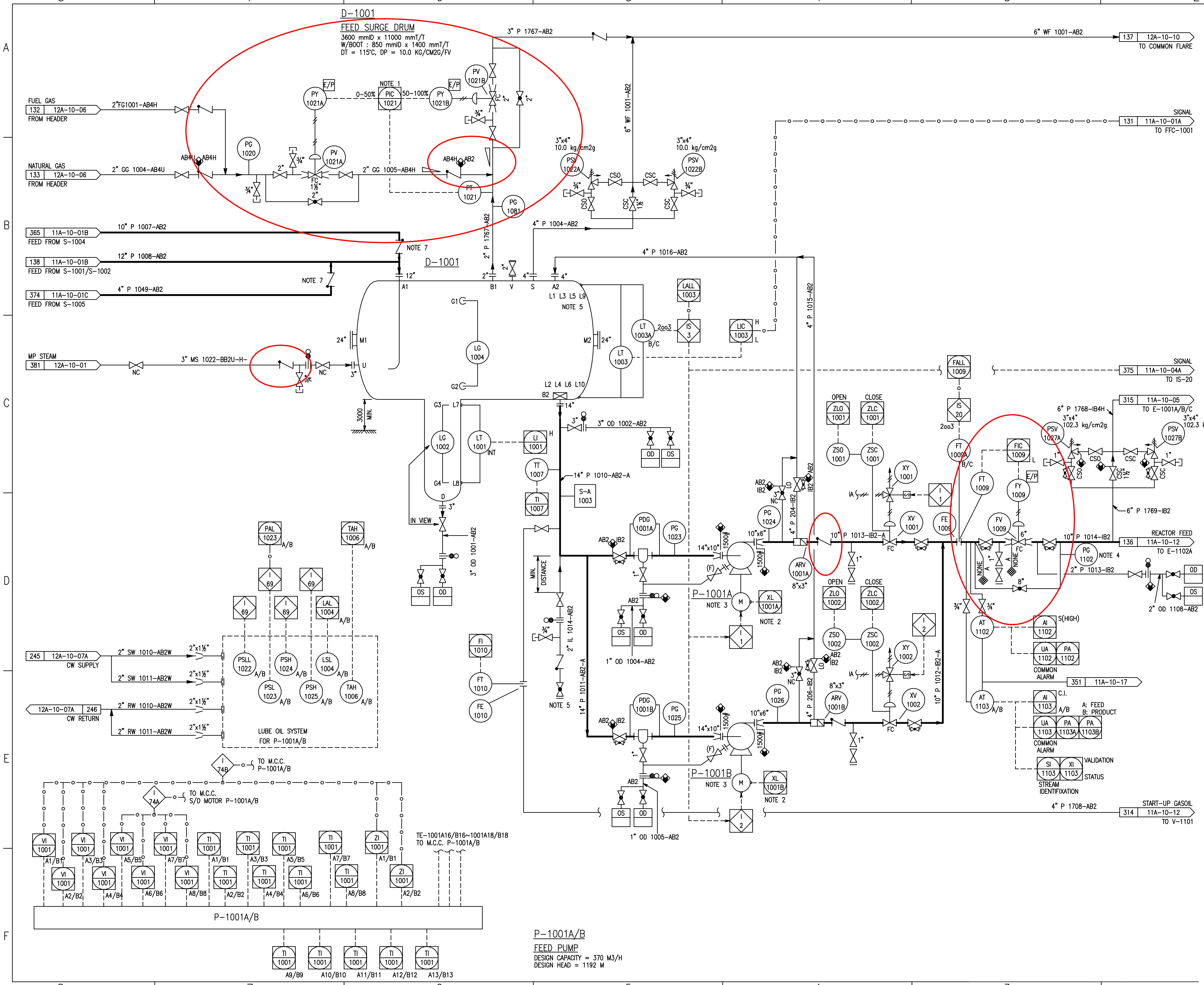
TYPE	DETAIL	P&ID SYMBOL
COLD LIQUID BELOW THE BOILING POINT		S A
LIQUID ABOVE 80°C OR ABOVE BP. LOW VISCOSITY. LOW POUR POINT.		SC B
LIQUID ABOVE 80°C HIGH VISCOSITY OR HIGH POUR POINT		SC C
REACTOR OUTLET		SC G

SAMPLE STATIONS

TYPE	DETAIL	P&ID SYMBOL
GAS, LPG AND TOXIC GAS BELOW 80°C		S D
GAS, LPG AND TOXIC GAS ABOVE 80°C		SC E

NOTES:

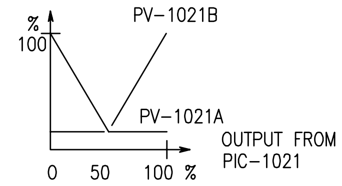
1. WHEN THE VALVE "A" IS ACCESSIBLE FROM THE SAME LOCATION AS VALVE "C". THE VALVE "B" IS NOT REQUIRED.
2. THE SAMPLING BOX IS REQUIRED ONLY FOR CLOSE SYSTEM WITH SAMPLING GLASS BOTTLE INSIDE IT.
   FOR OPEN SYSTEM, THE SAMPLING BOX IS NOT REQUIRED.
3. THE CHECK VALVE SHALL BE INSTALLED NEAR AND HIGH ABOVE FLARE HEADER.
4. REFER TO DWG NO. XB11A-0000-11.
5. THE SC-1004 & SC-1011 SHALL BE VENTED TO SAFE LOCATION, AND CHECK VALVE IS NOT REQUIRED.



NOTES:

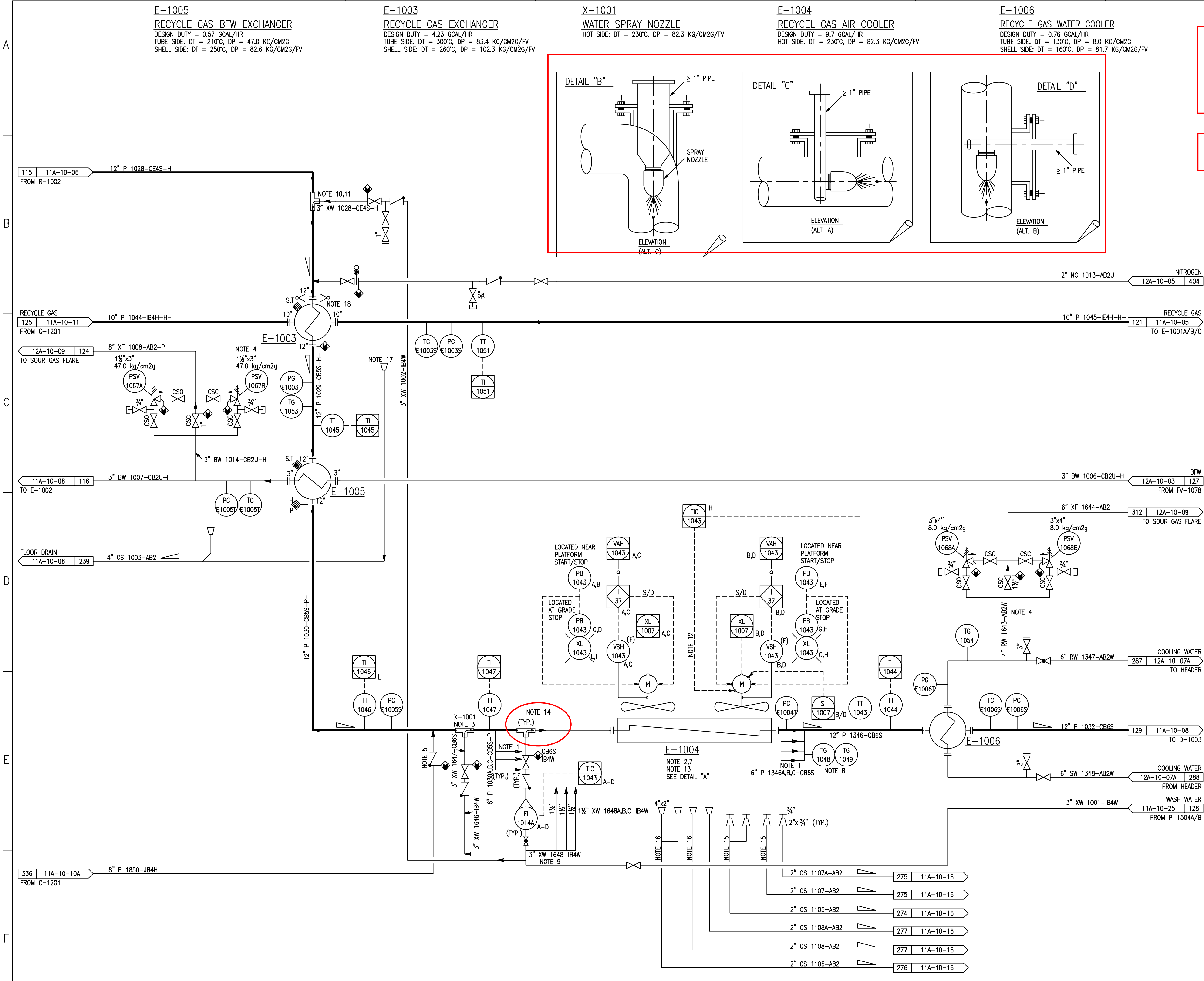
1. FUNCTION OF PIC-1021

VALVE OPENING



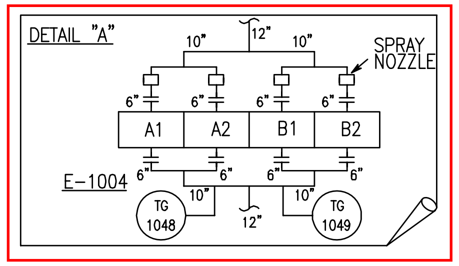
2. FOR MOTOR CONTROL SEE DWG. XB11A-0000-06 DETAIL "C".
3. THE EMERGENCY SHUTDOWN PUSH BUTTON SHALL BE LOCATED AT LEAST 15 M FROM PUMP MOTOR.
4. PG-1102 FOR CHECK E-1102A/B/C1/C2 OPERATE PRESSURE DROP.
5. FOR CATALYST SULPHIDING.
6. NOZZLE L1~L6 FOR LT-1003A/B/C(ESD), NOZZLE L9~L10 FOR LT-1003(DCS).
7. DISTANCE FROM CHECK VALVE TO TIE-IN POINT TO BE MINIMISED.

Att-2 (10/12)



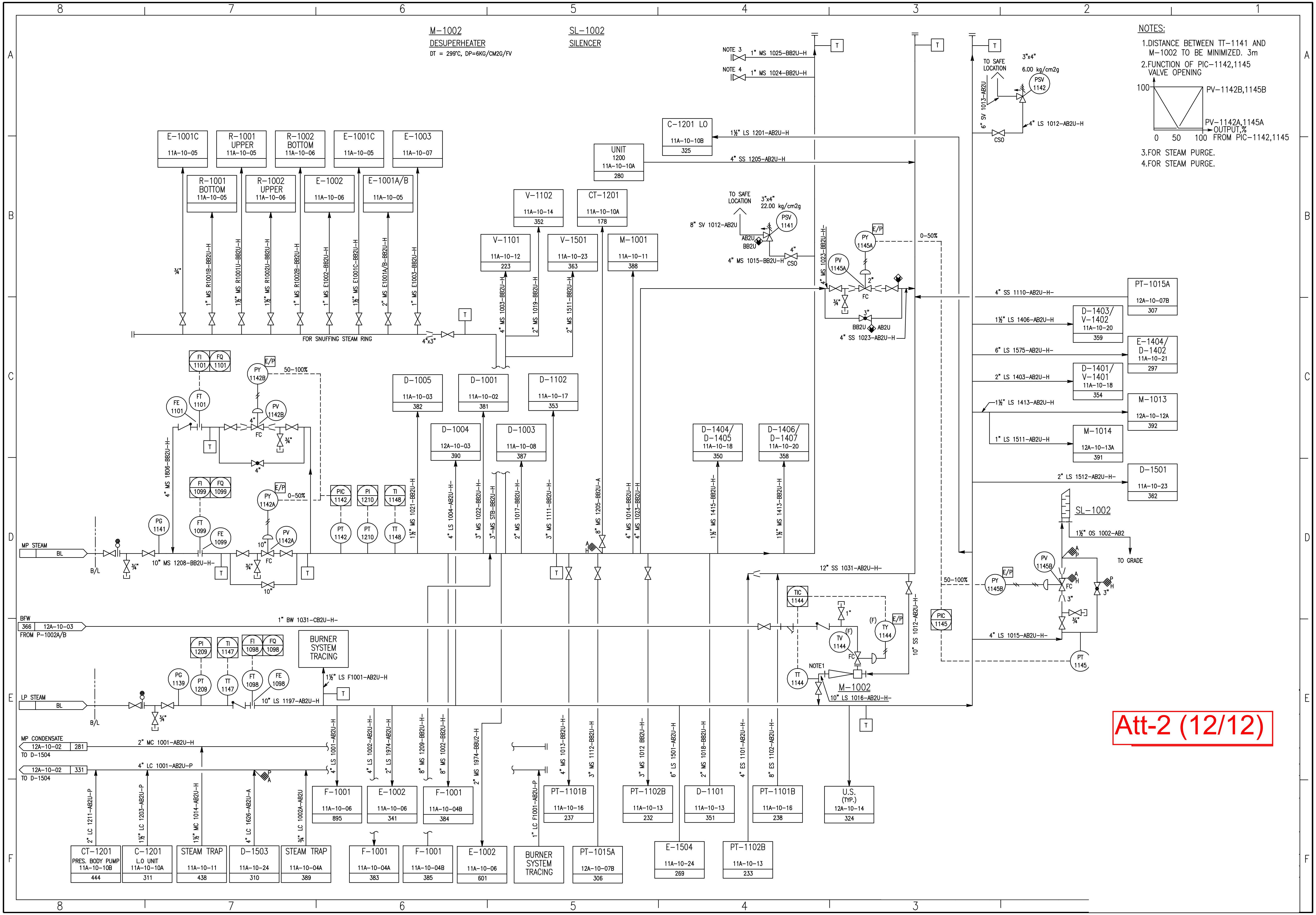
NOTES:

1. INLET AND OUTLET MANIFOLD TO BE SYMMETRICAL AND BALANCED, PIPING SPLIT TO BE NO BENDS 10D UPSTREAM OF SPLIT.
2. NUMBER OF BUNDLES TO BE VERIFIED BY VENDOR.
3. WATER SPRAY NOZZLE TO BE INSTALLED AT A MINIMUM DISTANCE OF 10 x PIPE ID TO DOWNSTREAM BEND.
4. TO BE LOCATED AT A MINIMUM DISTANCE FROM THE HEAT EXCHANGER NOZZLE.
5. DISTANCE FROM CHECK VALVE TO TIE-IN POINT TO BE MINIMISED.
6. DELETED.
7. DELETED.
8. TG-1048/1049 ARE LOCATED ON OUTLET OF EACH BAY. REFER TO DETAIL "A".
9. PIPING LAYOUT FOR EQUAL DISTRIBUTION.
10. WATER INJECTION LOCATED 1 M BELOW UPPER PIPE NECK.
11. DELETED.
12. TWO 4∼20mA SIGNALS.
13. FOR MOTOR CONTROL SEE DWG. NO. XB11A-0000-06 DETAIL "C".



14. WATER SPRAY NOZZLE TO BE INSTALLED AT A MINIMUM DISTANCE OF 5 x PIPE ID TO DOWNSTREAM BEND. LOCATE SPRAY NOZZLE IN HORIZONTAL OR VERTICAL SECTION OF VAPOR LINE TO AIR COOLER. SEE DETAIL "B" TO "D" FOR DETAIL.
15. FOR E-1004 GEAR BOX OIL DRAIN TO 50 GAL TANK.
16. FOR E-1004 OIL PAN TO D.F.
17. FOR HYDRAULIC TEST DRAINAGE NEAR E-1003/ E-1005.
18. :STEAM RING. DETAIL DRAWING SHALL BE REFERRED TO XB11A-0000-06. DETAIL DISTRIBUTION DRAWING SHALL BE REFERRED TO XB12A-1000-01.

Att-2 (11/12)


M-1002
DESUPERHEATER
DT = 299°C, DP=6KG/CM2G/FV
SL-1002
SILENCER
NOTES:
1.DISTANCE BETWEEN TT-1141 AND M-1002 TO BE MINIMIZED. 3m
2.FUNCTION OF PIC-1142,1145 VALVE OPENING
100
PV-1142B,1145B
PV-1142A,1145A
OUTPUT,%
0 50 100 FROM PIC-1142,1145
3.FOR STEAM PURGE.
4.FOR STEAM PURGE.
NOTE 3
1" MS 1025-BB2U-H
NOTE 4
1" MS 1024-BB2U-H
TO SAFE LOCATION
3"x4"
6.00 kg/cm2g
PSV 1142
6" SV 1013-AB2U
4" LS 1012-AB2U-H
CSO
E-1001C 11A-10-05
R-1001 UPPER 11A-10-05
R-1002 BOTTOM 11A-10-06
E-1001C 11A-10-05
E-1003 11A-10-07
R-1001 BOTTOM 11A-10-05
R-1002 UPPER 11A-10-06
E-1002 11A-10-06
E-1001A/B 11A-10-05
C-1201 LO 11A-10-10B 325
1½" LS 1201-AB2U-H
UNIT 1200 11A-10-10A 280
4" SS 1205-AB2U-H
V-1102 11A-10-14 352
CT-1201 11A-10-10A 178
V-1101 11A-10-12 223
V-1501 11A-10-23 363
M-1001 11A-10-11 388
TO SAFE LOCATION
3"x4"
22.00 kg/cm2g
PSV 1141
8" SV 1012-AB2U
4" MS 1015-BB2U-H
CSO
PY 1145A
PV 1145A
E/P
0-50%
FC
4" SS 1023-BB2U-H
4" SS 1110-AB2U-H-
PT-1015A 12A-10-07B 307
1½" LS 1406-AB2U-H
D-1403/ V-1402 11A-10-20 359
6" LS 1575-AB2U-H-
E-1404/ D-1402 11A-10-21 297
2" LS 1403-AB2U-H
D-1401/ V-1401 11A-10-18 354
1½" LS 1413-AB2U-H
M-1013 12A-10-12A 392
1" LS 1511-AB2U-H
M-1014 12A-10-13A 391
2" LS 1512-AB2U-H-
D-1501 11A-10-23 362
¾"
1" MS R1001B-BB2U-H
1½" MS R1001U-BB2U-H
1½" MS R1002U-BB2U-H
1" MS R1002B-BB2U-H
1" MS E1002-BB2U-H
1½" MS E1001C-BB2U-H
2" MS E1001A/B-BB2U-H
1" MS E1003-BB2U-H
FOR SNUFFING STEAM RING
4"x3"
4" MS 1003-BB2U-H
2" MS 1019-BB2U-H
2" MS 1511-BB2U-H
FI 1101
FQ 1101
PY 1142B
E/P
50-100%
FE 1101
FT 1101
PV 1142B
FC
D-1005 11A-10-03 382
D-1001 11A-10-02 381
D-1102 11A-10-17 353
D-1004 12A-10-03 390
D-1003 11A-10-08 387
D-1404/ D-1405 11A-10-18 350
D-1406/ D-1407 11A-10-20 358
4" MS 1806-BB2U-H
FI 1099
FQ 1099
PY 1142A
E/P
0-50%
PIC 1142
PI 1210
TI 1148
PG 1141
FT 1099
FE 1099
PV 1142A
PT 1142
PT 1210
TT 1148
MP STEAM
BL
B/L
10" MS 1208-BB2U-H-
FC
1½" MS 1021-BB2U-H
4" LS 1004-AB2U-H-
3" MS 1022-BB2U-H
3" MS STB-BB2U-H
2" MS 1017-BB2U-H
3" MS 1111-BB2U-H
8" MS 1205-BB2U-A
4" MS 1014-BB2U-H
4" MS 1023-BB2U-H
1½" MS 1415-BB2U-H-
1½" MS 1413-BB2U-H
SL-1002
1½" OS 1002-AB2
TO GRADE
12" SS 1031-AB2U-H-
50-100%
PY 1145B
E/P
PV 1145B
FC
TIC 1144
TY 1144
E/P
TV 1144
PIC 1145
10" SS 1012-AB2U-H-
4" LS 1015-AB2U-H-
PT 1145
NOTE1
TT 1144
M-1002
10" LS 1016-AB2U-H-
BFW
366 12A-10-03
FROM P-1002A/B
1" BW 1031-CB2U-H-
PI 1209
TI 1147
FI 1098
FQ 1098
BURNER SYSTEM TRACING
PG 1139
PT 1209
TT 1147
FT 1098
FE 1098
1½" LS F1001-AB2U-H
LP STEAM
BL
B/L
10" LS 1197-AB2U-H
MP CONDENSATE
12A-10-02 281
TO D-1504
2" MC 1001-AB2U-H
12A-10-02 331
TO D-1504
4" LC 1001-AB2U-P
4" LS 1001-AB2U-H
4" LS 1002-AB2U-H-
2" LS 1974-AB2U-H
8" MS 1209-BB2U-H-
8" MS 1002-BB2U-H
2" MS 1974-BB2U-H
1" LC F1001-AB2U-P
4" MS 1013-BB2U-H-
3" MS 1112-BB2U-H
3" MS 1012 BB2U-H-
6" LS 1501-AB2U-H
2" MS 1018-BB2U-H
4" ES 1101-AB2U-H-
8" ES 1102-AB2U-H-
2" LC 1211-AB2U-P
1½" LC 1203-AB2U-P
1½" MC 1014-AB2U-H
4" LC 1626-AB2U-A
¾" LC 1002A-AB2U
F-1001 11A-10-06 895
E-1002 11A-10-06 341
F-1001 11A-10-04B 384
PT-1101B 11A-10-16 237
PT-1102B 11A-10-13 232
D-1101 11A-10-13 351
PT-1101B 11A-10-16 238
U.S. (TYP.) 12A-10-14 324
CT-1201 PRES. BODY PUMP 11A-10-10B 444
C-1201 L.O UNIT 11A-10-10A 311
STEAM TRAP 11A-10-11 438
D-1503 11A-10-24 310
STEAM TRAP 11A-10-04A 389
F-1001 11A-10-04A 383
F-1001 11A-10-04B 385
E-1002 11A-10-06 601
BURNER SYSTEM TRACING
PT-1015A 12A-10-07B 306
E-1504 11A-10-24 269
PT-1102B 11A-10-13 233

Att-2 (12/12)